LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - I

Matrices

Topics covered:

- Rank of a matrix: echelon form and normal form
- Cauchy-Binet formula (without proof)
- Inverse of a non-singular matrix by Gauss-Jordan method
- Systems of linear equations - homogeneous and non-homogeneous
- Gauss elimination method
- Jacobi and Gauss-Seidel iteration methods

Handwritten Notes - B.Tech (JNTUA R23)

1. Rank of a Matrix

Definition: the rank $r(A)$ of an $m \times n$ matrix A is the order of the largest non-zero minor (determinant) of A. Equivalently it is the number of NON-ZERO ROWS in its row echelon form.

- $r(A) = r(A^T)$, $r(A) \le \min(m, n)$, $r(A) = 0$ only for the zero matrix.
- For a square matrix of order n: $r(A) = n \Leftrightarrow |A| \neq 0$ (non-singular).
- Elementary row / column operations do NOT change the rank.
- $r(AB) \le \min(r(A), r(B))$.

Echelon form: a matrix is in (row) echelon form if (i) all zero rows are at the bottom, (ii) the first non-zero entry of each row (leading entry) lies to the right of the leading entry of the row above, so entries below a leading entry are zero.

Elementary operations: interchange $R_i \leftrightarrow R_j$; multiply a row by a non-zero constant $R_i \to kR_i$; add a multiple of one row to another $R_i \to R_i + kR_j$. (The same with columns C_i.)

Method: use the 1st row to make zeros below the first leading entry, then the 2nd row for the next column, and so on. Count the non-zero rows.

Problem 1: find the rank of $A = [1\ 2\ -1\ 3;\ 2\ 4\ 1\ -2;\ 3\ 6\ 3\ -7]$.

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 3 & 6 & 3 & -7 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 6 & -16 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Two non-zero rows remain $\Rightarrow r(A) = 2$.

Problem 2: find the rank of $A = [1\ 2\ 3;\ 2\ 4\ 7;\ 3\ 6\ 10]$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 3 & 6 & 10 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence $r(A) = 2$. (Check: $|A| = 0$ but the minor of order 2 formed by columns 1, 3 and rows 1, 2 is $1 \cdot 7 - 3 \cdot 2 = 1 \neq 0$.)

Problem 3: find the rank of the 4 x 4 matrix $A = [1\ 2\ 3\ 4;\ 2\ 3\ 4\ 5;\ 3\ 4\ 5\ 6;\ 4\ 5\ 6\ 7]$.

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & -2 & -4 & -6 \\ 4 & 5 & 6 & 7 \end{bmatrix}$$

$$\xrightarrow{R_4 \to R_4 - 4R_1} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & -2 & -4 & -6 \\ 0 & -3 & -6 & -9 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & -6 & -9 \end{bmatrix} \xrightarrow{R_4 \to R_4 - 3R_2} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$r(A) = 2$ (the rows are in arithmetic progression, so every row is a combination of the first two).

Problem 4: find the rank of $A = [2\ 3\ -1\ -1;\ 1\ -1\ -2\ -4;\ 3\ 1\ 3\ -2;\ 6\ 3\ 0\ -7]$.

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 6 & 3 & 0 & -7 \end{bmatrix} \xrightarrow{R_4 \to R_4 - 6R_1} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix} \xrightarrow{R_3 \to 5R_3 - 4R_2} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 33 & 22 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$$\xrightarrow[\sim]{R4\to 5R4-9R2}\begin{bmatrix}1&-1&-2&-4\\0&5&3&7\\0&0&33&22\\0&0&33&22\end{bmatrix}\xrightarrow[\sim]{R4\to R4-R3}\begin{bmatrix}1&-1&-2&-4\\0&5&3&7\\0&0&33&22\\0&0&0&0\end{bmatrix}$$

Three non-zero rows (the last row has become zero) $\Rightarrow r(A) = 3$. The matrix is singular since $r < 4$.

Problem 5: find the rank of $A = [1\ 2\ 3;\ 2\ 4\ \lambda;\ 3\ 6\ 9]$ for different values of λ.

$$A=\begin{bmatrix}1&2&3\\2&4&\lambda\\3&6&9\end{bmatrix}\xrightarrow[\sim]{R2-2R1,\ R3-3R1}\begin{bmatrix}1&2&3\\0&0&\lambda-6\\0&0&0\end{bmatrix}$$

If $\lambda \neq 6$ the second row is non-zero $\Rightarrow r(A) = 2$. If $\lambda = 6$ the second row also vanishes $\Rightarrow r(A) = 1$.

2. Normal Form (Canonical Form)

Every $m \times n$ matrix of rank r can be reduced, by elementary ROW and COLUMN operations, to the normal form

$N = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, and then $r(A) = r$. Equivalently $PAQ = N$ with P, Q non-singular.

Problem 6: reduce $A = [1\ 2\ 3\ 2;\ 2\ 3\ 5\ 1;\ 1\ 3\ 4\ 5]$ to normal form and find its rank.

$$A=\begin{bmatrix}1&2&3&2\\2&3&5&1\\1&3&4&5\end{bmatrix}\xrightarrow[\sim]{R2\to R2-2R1}\begin{bmatrix}1&2&3&2\\0&-1&-1&-3\\1&3&4&5\end{bmatrix}\xrightarrow[\sim]{R3\to R3-R1}\begin{bmatrix}1&2&3&2\\0&-1&-1&-3\\0&1&1&3\end{bmatrix}$$

$$\xrightarrow[\sim]{R3\to R3+R2}\begin{bmatrix}1&2&3&2\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}\xrightarrow[\sim]{C2\to C2-2C1}\begin{bmatrix}1&0&3&2\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}\xrightarrow[\sim]{C3\to C3-3C1}\begin{bmatrix}1&0&0&2\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}$$

$$\xrightarrow{C4\to C4-2C1}\sim\begin{bmatrix}1&0&0&0\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}\xrightarrow{R2\to -R2}\sim\begin{bmatrix}1&0&0&0\\0&1&1&3\\0&0&0&0\end{bmatrix}\xrightarrow{C3\to C3-C2}\sim\begin{bmatrix}1&0&0&0\\0&1&0&3\\0&0&0&0\end{bmatrix}$$

$$\xrightarrow{C4\to C4-3C2}\sim\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&0\end{bmatrix}$$

This is $[I_2\ 0\ ;\ 0\ 0]\Rightarrow r(A)=2$.

Problem 7: find non-singular matrices P and Q such that PAQ is in normal form, for $A=[1\,1\,1\,;\,1\,2\,3\,;\,2\,3\,4]$.

Write $A=I\,A\,I$ and perform every row operation also on the LEFT identity and every column operation on the RIGHT identity:

$$R2-R1,\ R3-2R1,\ R3-R2\ ;\ C2-C1,\ C3-C1,\ C3-2C2$$

$$P=\begin{bmatrix}1&0&0\\-1&1&0\\-1&-1&1\end{bmatrix}\quad Q=\begin{bmatrix}1&-1&1\\0&1&-2\\0&0&1\end{bmatrix}\quad PAQ=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$$

So $PAQ=[I_2\ 0\ ;\ 0\ 0]$ and $r(A)=2$.

3. Cauchy–Binet Formula (without proof)

Let A be an $m\times n$ matrix and B an $n\times m$ matrix. Then $\det(AB)$ is the sum, over all choices S of m columns of A (equivalently the same m rows of B), of the product of the corresponding minors:

$$\boxed{\det(AB)=\Sigma\ \det(A_S)\cdot\det(B_S)}$$

(the sum is over all $C(n,m)$ selections S of m columns).

- If $m\le n$: $A_S=m\times m$ sub-matrix of A formed by the columns in S; $B_S=m\times m$ sub-matrix of B formed by the same rows.
- If $m>n$: $\det(AB)=0$ (because $r(AB)\le n<m$).
- If $m=n$ it reduces to $\det(AB)=\det A\cdot\det B$.

Problem 8: verify the formula for $A = [1\ 2\ 3; 4\ 5\ 6]$ and $B = [1\ 0; 0\ 1; 1\ 1]$.

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 10 & 11 \end{bmatrix} \Rightarrow \det(AB) = -6$$

columns 1,2: $\det A_S = -3$, $\det B_S = 1 \rightarrow$ product $= -3$

columns 1,3: $\det A_S = -6$, $\det B_S = 1 \rightarrow$ product $= -6$

columns 2,3: $\det A_S = -3$, $\det B_S = -1 \rightarrow$ product $= 3$

Sum $= -6 = \det(AB)$ ✓ Cauchy–Binet verified.

Problem 9: for $A = [1\ 2; 3\ 4; 5\ 6]$ (3 × 2) and $B = [1\ 0\ 1; 2\ 1\ 0]$ (2 × 3) find $\det(AB)$.

Here $m = 3 > n = 2$, so by Cauchy–Binet $\det(AB) = 0$. (Direct check: AB is 3 × 3 of rank ≤ 2, hence singular.)

4. Inverse by Gauss–Jordan Method

A^{-1} exists only if $|A| \neq 0$. Write the augmented matrix $[A\,|\,I]$ and apply row operations ONLY until the left half becomes I; the right half is then A^{-1}. $[A\,|\,I] \sim [I\,|\,A^{-1}]$.

If a zero row appears in the left half, A is singular – no inverse.

Problem 10: find the inverse of $A = [1\ 2\ 3; 2\ 5\ 3; 1\ 0\ 8]$.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 3R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 + 3R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 2R_2} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

Check: $A \cdot A^{-1} = I$ ✓.

Problem II: find the inverse of $A = [1\,1\,1;\ 1\,2\,3;\ 1\,4\,9]$.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 4 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 1 & 4 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_1} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 3 & 8 & -1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 3R_2} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 2 & -3 & 1 \end{array}\right] \xrightarrow{R_3 \to (1/2)R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 2R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 3/2 & -1/2 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -5/2 & 1/2 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 3 & -5/2 & 1/2 \\ -3 & 4 & -1 \\ 1 & -3/2 & 1/2 \end{bmatrix}$$

Problem 12: is $A = [1\ 2\ ;\ 2\ 4]$ invertible?

$$[A|I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right]$$

A zero row appears in the left half ($|A| = 4 - 4 = 0$) $\Rightarrow$ A is singular, A^{-1} does not exist.

5. System of Linear Equations

$AX = B$ (m equations, n unknowns). Augmented matrix $[A \mid B] = [A\ B]$. Reduce it to echelon form (Gauss elimination). Rouché–Capelli test:

Condition	Nature of solution
$r(A) \neq r([A\|B])$	INCONSISTENT – no solution
$r(A) = r([A\|B]) = n$ (no. of unknowns)	CONSISTENT – UNIQUE solution
$r(A) = r([A\|B]) = r < n$	CONSISTENT – INFINITELY many solutions ($n - r$ free variables)

Homogeneous system $AX = 0$ is always consistent ($X = 0$ trivial solution). Non-trivial solutions exist $\Leftrightarrow$ $r(A) < n$ (for a square system $\Leftrightarrow |A| = 0$).

Gauss elimination: convert $[A \mid B]$ to upper-triangular (echelon) form by row operations, then solve by BACK-SUBSTITUTION.

Problem 13 (unique solution): $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + (-1/2)R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -3 & -9 \end{array}\right]$$

$r(A) = r([A|B]) = 3 = n \Rightarrow$ unique solution. Back-substitution: $-3z = -9 \to z = 3$; $-2y = -4 \to y = 2$; $x + 2 + 3 = 6 \to x = 1$.

$$\boxed{x = 1, y = 2, z = 3}$$

Problem 14 (infinitely many): $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 3y + 4z = 20$.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 20 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 2 & 3 & 4 & 20 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & 2 & 8 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$r(A) = r([A|B]) = 2 < 3 = n \Rightarrow$ infinitely many solutions (one free variable). Put $z = K$: $y + 2z = 8 \to y = 8 - 2K$; $x = 6 - y - z = K - 2$.

$$\boxed{x = K - 2, y = 8 - 2K, z = K \ (K \text{ any real number})}$$

Problem 15 (inconsistent): $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 3y + 4z = 21$.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 21 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 2 & 3 & 4 & 21 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & 2 & 9 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$r(A) = 2$ but $r([A|B]) = 3$ (last row reads $0 = 1$) $\Rightarrow$ the system is INCONSISTENT – no solution.

Problem 16: for what values of λ and μ does $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have (i) no solution (ii) a unique solution (iii) infinitely many solutions?

$$[A|B] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right] \quad (R_2-R_1,\ R_3-R_1,\ R_3-R_2)$$

- $\lambda \neq 3$: $r(A) = r([A|B]) = 3 \rightarrow$ UNIQUE solution (for any μ).
- $\lambda = 3$ and $\mu \neq 10$: $r(A) = 2$, $r([A|B]) = 3 \rightarrow$ NO solution.
- $\lambda = 3$ and $\mu = 10$: $r(A) = r([A|B]) = 2 \rightarrow$ INFINITELY many solutions.

Problem 17 (homogeneous): solve $x + 3y - 2z = 0$, $2x - y + 4z = 0$, $x - 11y + 14z = 0$.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 1 & -11 & 14 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$r(A) = 2 < 3 \Rightarrow$ non-trivial solutions. Put $z = K$: $-7y + 8K = 0 \rightarrow y = 8K/7$; $x = -3y + 2z = -10K/7$. Taking $K = 7t$:

$$\boxed{(x, y, z) = t(-10, 8, 7)}$$

Problem 18: find λ for which $x + 2y + 3z = \lambda x$, $3x + y + 2z = \lambda y$, $2x + 3y + z = \lambda z$ has a non-trivial solution; solve it.

Non-trivial solution exists iff $|A - \lambda I| = 0$: expanding, $\lambda^3 - 3\lambda^2 - 9\lambda - 18 = 0$ i.e. $(\lambda - 6)(\lambda^2 + 3\lambda + 3) = 0$.

The only real value is $\lambda = 6$. For $\lambda = 6$: $-5x + 2y + 3z = 0$, $3x - 5y + 2z = 0$. By cross-multiplication $x : y : z = (4 + 15) : (9 + 10) : (25 - 6) = 19 : 19 : 19 \rightarrow x = y = z = K$.

$$\boxed{\lambda = 6,\ x = y = z = K}$$

6. Jacobi and Gauss-Seidel Iteration Methods

Used for large systems. The system must first be arranged so that the coefficient matrix is DIAGONALLY DOMINANT: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ (each diagonal element larger than the sum of the others in its row). Then both methods converge for any starting value.

For $a_1x + b_1y + c_1z = d_1$, etc. solve the i-th equation for the i-th unkno

$$\text{Jacobi}: x^{(K+1)} = \frac{d_1 - b_1y^{(K)} - c_1z^{(K)}}{a_1}, \; y^{(K+1)} \text{ uses } x^{(K)}, z^{(K)}, \ldots$$

$$\text{Gauss-Seidel}: y^{(K+1)} = \frac{d_2 - a_2x^{(K+1)} - c_2z^{(K)}}{b_2}, \ldots$$

(i.e. the NEWEST available values are used at once.)

Jacobi = simultaneous displacement (all old values); Gauss-Seidel = successive displacement (new values used immediately) – converges about twice as fast.

Problem 19: solve $10x + y + z = 12$, $2x + 10y + z = 13$, $2x + 2y + 10z = 14$ by Jacobi's method (start with $x = y = z = 0$).

Diagonally dominant ($10 > 2$, $10 > 3$, $10 > 4$) ✓. Iteration:

$$x = (12 - y - z)/10, \; y = (13 - 2x - z)/10, \; z = (14 - 2x - 2y)/10$$

K	x	y	z
0	0.0000	0.0000	0.0000
1	1.2000	1.3000	1.4000
2	0.9300	0.9200	0.9000
3	1.0180	1.0240	1.0300
4	0.9946	0.9934	0.9916
5	1.0015	1.0019	1.0024
6	0.9996	0.9995	0.9993

The values approach $x = y = z = 1$. (Exact solution 1, 1, 1.)

Problem 20: solve the same system by the Gauss-Seidel method.

K	x	y	z
0	0.0000	0.0000	0.0000
1	1.2000	1.0600	0.9480
2	0.9992	1.0054	0.9991
3	0.9996	1.0002	1.0001
4	1.0000	1.0000	1.0000

Already after 4 iterations Gauss-Seidel is almost at (1, 1, 1) – faster than Jacobi.

Problem 21: solve $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$ by Gauss-Seidel.

$x = (17 - y + 2z)/20$, $y = (-18 - 3x + z)/20$, $z = (25 - 2x + 3y)/20$

K	x	y	z
0	0.0000	0.0000	0.0000
1	0.8500	-1.0275	1.0109
2	1.0025	-0.9998	0.9998
3	1.0000	-1.0000	1.0000
4	1.0000	-1.0000	1.0000

Solution: $x \approx 1.000$, $y \approx -1.000$, $z \approx 1.000$.

Problem 22: solve $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$ by Gauss-Seidel (4 iterations).

K	x	y	z
0	0.0000	0.0000	0.0000
1	3.1481	3.5407	1.9132
2	2.4322	3.5720	1.9258
3	2.4257	3.5729	1.9260
4	2.4255	3.5730	1.9260

$x = 2.4255$, $y = 3.5730$, $z = 1.9260$ (to 4 decimals).

Problem 23: rearrange $x + 2y + 10z = 24$, $10x + y + z = 12$, $2x + 10y + z = 13$ so that the method converges.

Interchange so that the large coefficient is on the diagonal: $10x + y + z = 12$, $2x + 10y + z = 13$, $x + 2y + 10z = 24$. Now $10 > 2$, $10 > 3$, $10 > 3$ and Jacobi / Gauss-Seidel can be applied as above.

Important Questions - Unit I

1. Reduce a matrix to echelon form and find its rank (problems 1-5).
2. Reduce a matrix to normal form; find P and Q such that PAQ = normal form.
3. State the Cauchy-Binet formula and verify it for given matrices.
4. Find the inverse of a matrix by the Gauss-Jordan method.
5. Test consistency and solve a system by Gauss elimination (unique / infinite / none).
6. Find λ, μ for which a system has no / unique / infinitely many solutions.
7. Solve a homogeneous system; find non-trivial solutions.
8. Solve a system by Jacobi's and Gauss-Seidel iteration (3-4 iterations).